

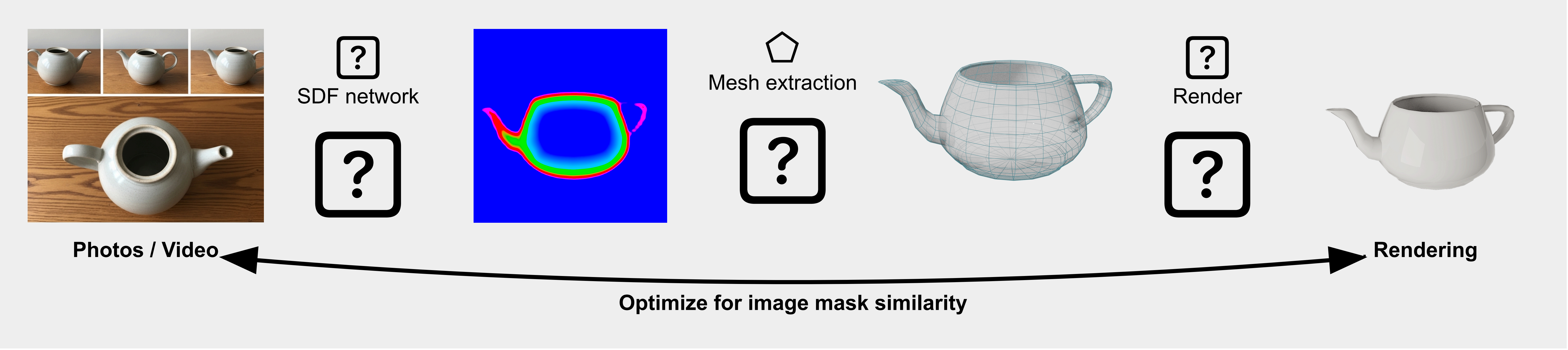
Mesh Extraction with FlexiCubes

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Mesh extraction pipeline



Deep Marching Tetraeda



Flexicubes



[1] Similar to Dual Marching cubes, flexicubes uses both the dual and primal grid. It does so by first sampling the SDF on the primal grid, creating interpolation weights α along grid edges. A primal mesh is created similar to marching cubes

Results

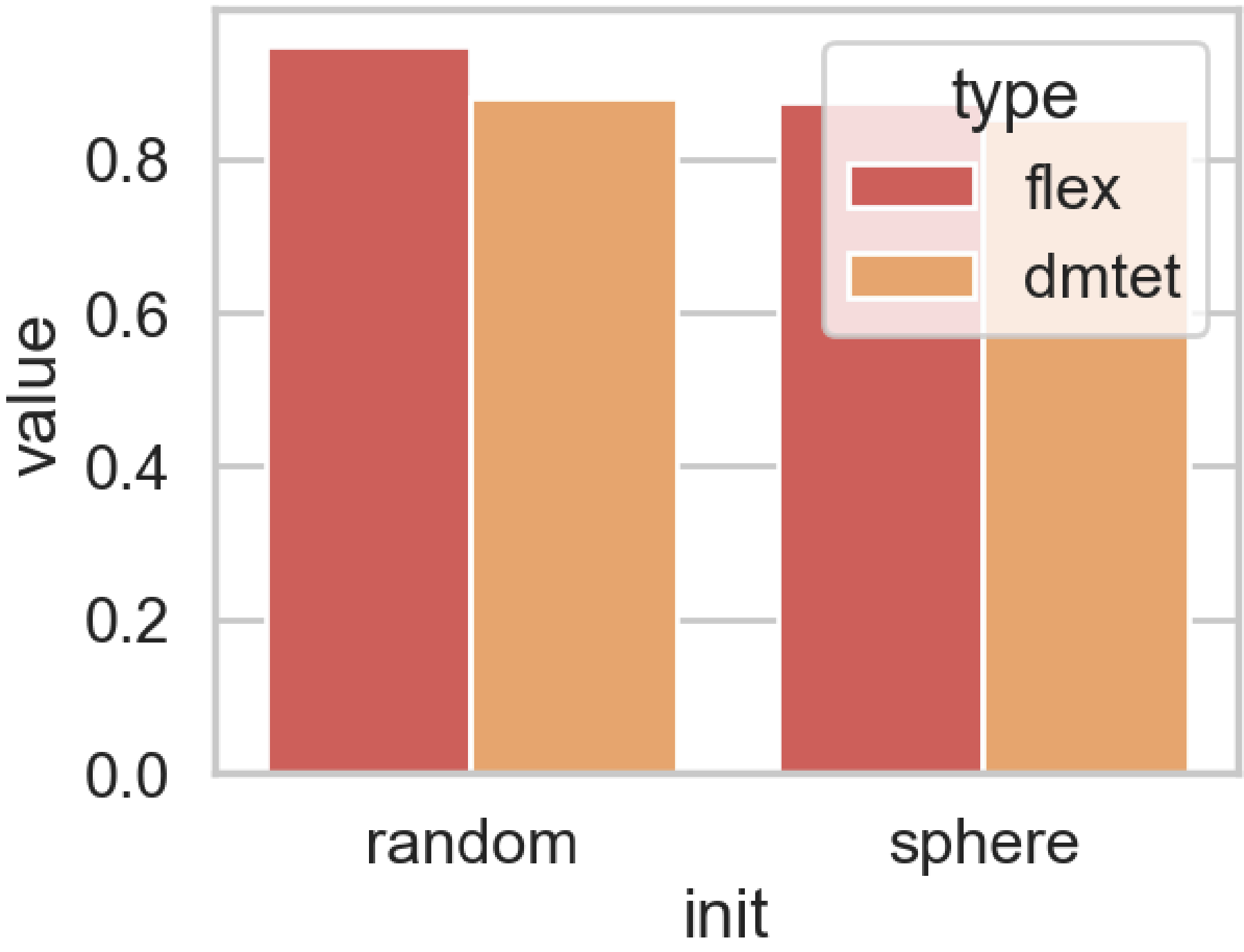
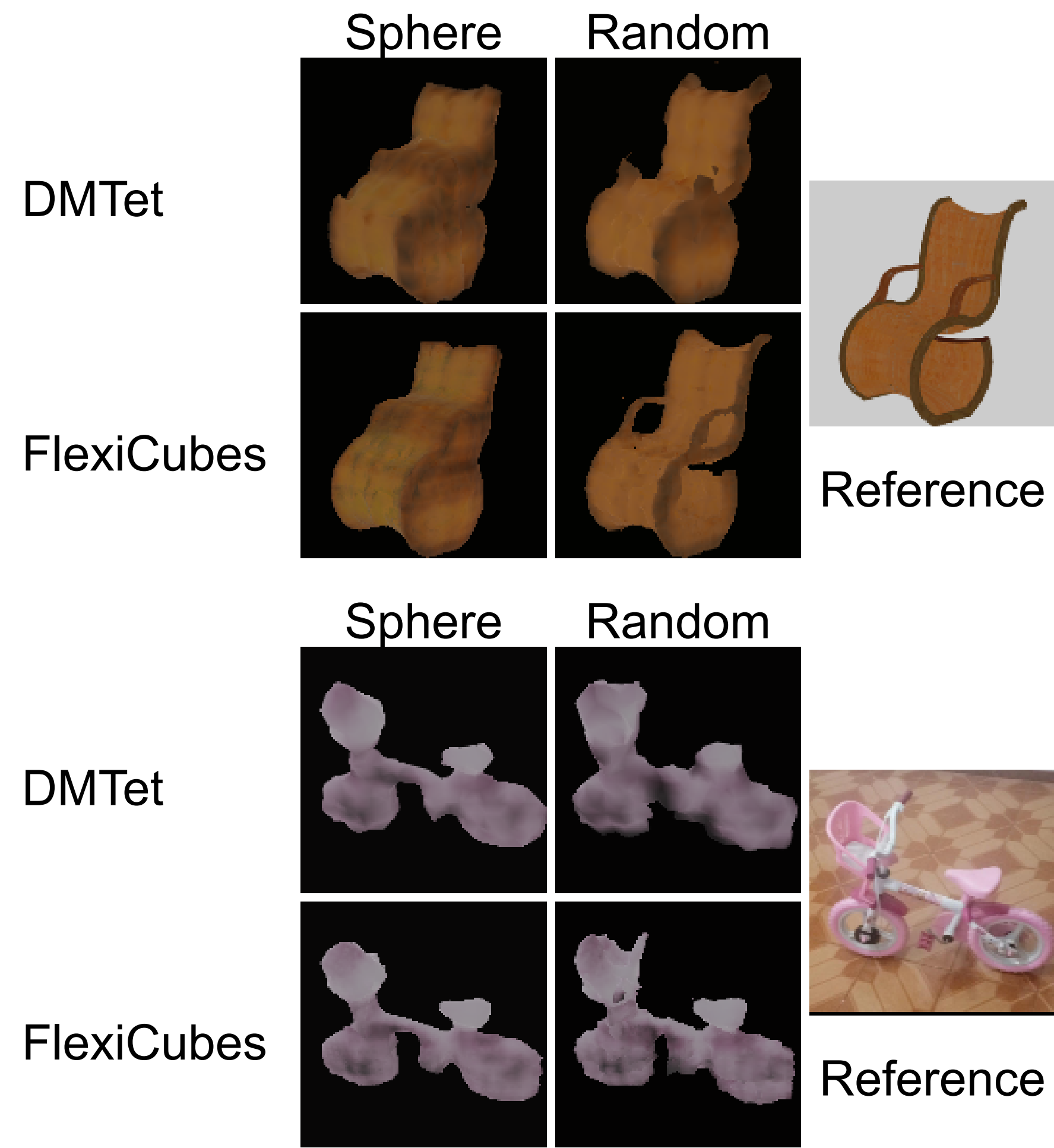


Figure 1: IoU at Test time (ShapeNet)

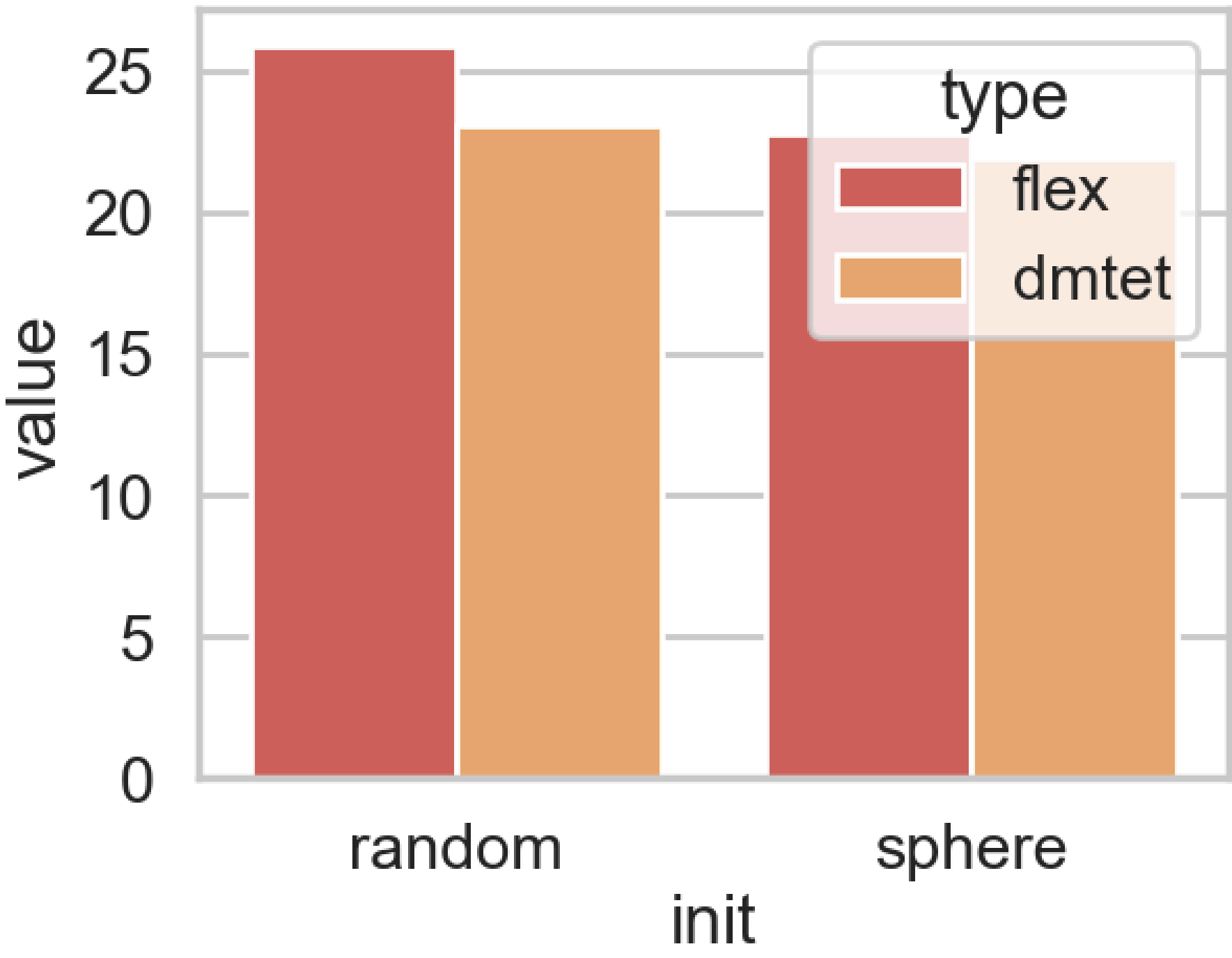


Figure 3: PSNR at Test time (ShapeNet)

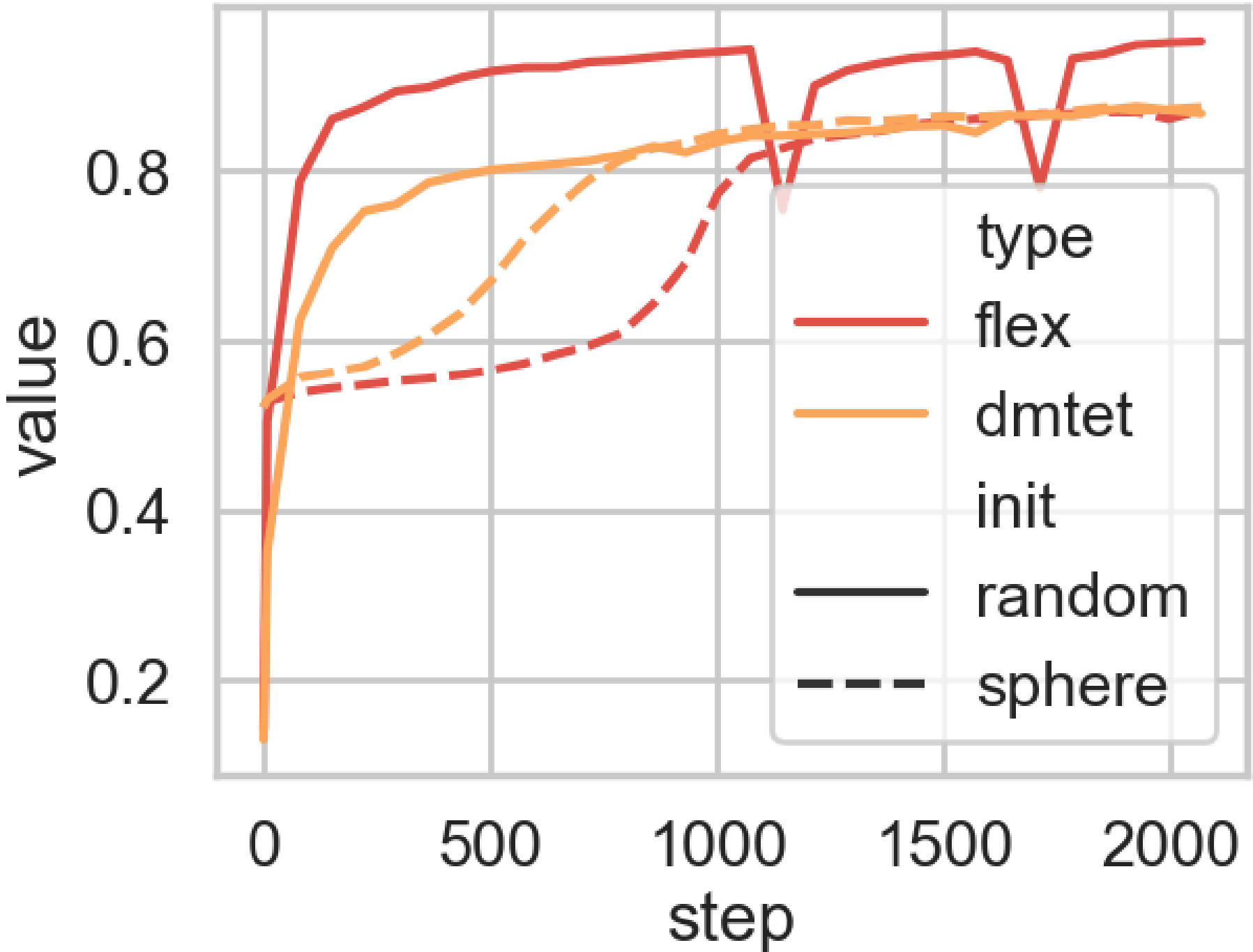


Figure 2: IoU on Validation Set during Training

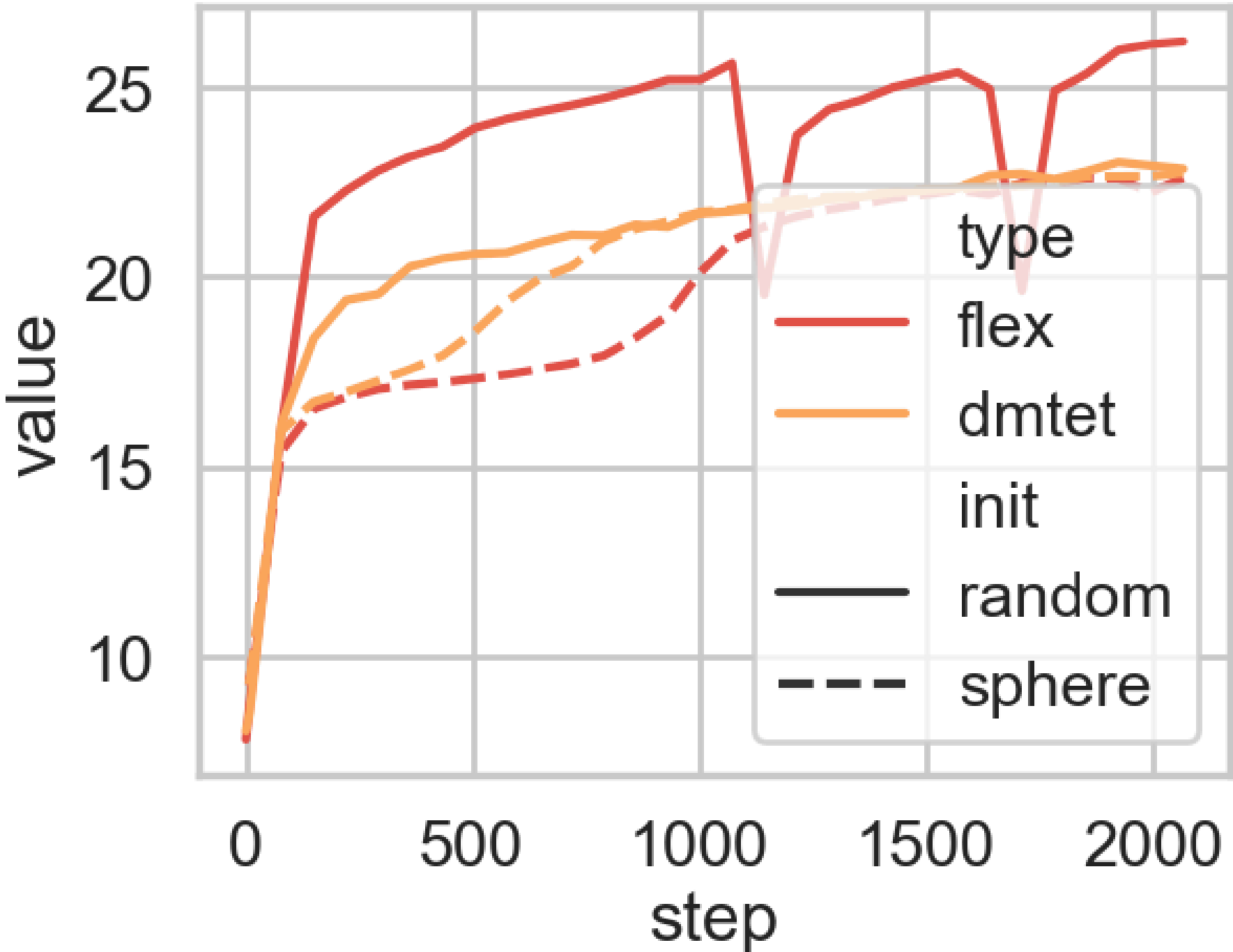


Figure 4: PSNR on Validation Set during Training

Bibliography

[1] T. Shen *et al.*, "Flexible Isosurface Extraction for Gradient-Based Mesh Optimization," *ACM Transactions on Graphics*, vol. 42, no. 4, pp. 1–16, Jul. 2023, doi: 10.1145/3592430.